Huawei Access Node HSR-ACC-64 Hardware/Environmental Fault SOP

Node: HSR-ACC-64 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_4101

Description

This Standard Operating Procedure (SOP) addresses a Major failure domain on Huawei Access Node HSR-ACC-64 (Role: Access), specifically related to Hardware/Environmental Faults. The failure domain is primarily triggered by ASIC memory parity errors, line card crashes, and power supply redundancy loss. This equipment is situated at the HSR Layout Access site and is operating under the Huawei CLI dialect.

Pre-requisite Checks

- Verify that remote hands are dispatched to the HSR Layout Access site to initiate physical checks and ensure access to the node.

- Check the status of redundant links to confirm proper operation and redundancy.

- Perform a visual inspection of the node's external and internal components to detect any signs of physical damage or degradation.

- Review the node's environmental operating parameters to ensure compliance with the manufacturer's specifications.

Potential Root Causes

Cause	Probability
ASIC memory parity error	High
Line card crash	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display current-configuration | include ASIC, Expected Output: System configuration data., Description: This command will display the current configuration of the ASIC, including any relevant settings or parameters that may be indicative of a parity error.

Command: show power supply, Expected Output: Power supply status information., Description: This command will provide information on the power supply status, including any alerts or warnings related to redundancy or failure.

Command: monitor interface LineCard, Expected Output: Interface LineCard monitoring data., Description: This command will display monitoring data for the line card, including any error or alert messages.
```

Resolution Steps

Step 1: Step: Replace ASIC module, Command: replace ASIC module, Description: Replace the faulty ASIC module with a new one, ensuring proper installation and configuration.

Step 2: Step: Reboot node, Command: reset system, Description: Reboot the node to ensure that any changes or repairs take effect.

Step 3: Step: Verify power supply redundancy, Command: show power supply, Description: Verify that the power supply redundancy is operational and functioning as expected.

Post-Check Verification

```
Command: display current-configuration | include ASIC, Expected Output: System configuration data., Description: Verify that the ASIC configuration has been updated successfully.

Command: show power supply, Expected Output: Power supply status information., Description: Verify that the power supply redundancy is operational and functioning as expected.

Command: monitor interface LineCard, Expected Output: Interface LineCard monitoring data., Description: Verify that the line card is functioning properly and there are no error or alert messages.
```

Roll-back Procedures

- Step: Restore original ASIC configuration, Command: undo replace ASIC module, Description: Restore the original ASIC configuration to the previous state in case the replacement fails.

- Step: Revert power supply configuration, Command: undo power supply configuration, Description: Revert the power supply configuration to the previous state in case the repair fails.